SCSI 1013 DISCRETE STRUCTURE

CHAPTER 4 – PART 2

TREE

Semester 1 – 2019/2020

Introduction

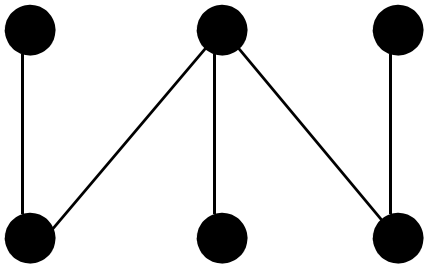
Definition 1. A tree is a connected undirected graph with no simple circuits.

Theorem 1. An undirected graph is a tree if and only if there is a unique simple path between any two of its vertices.

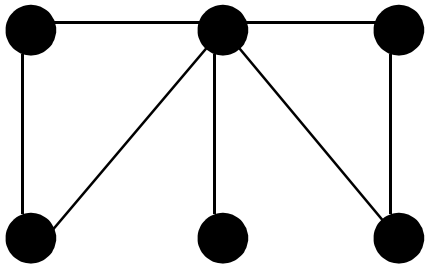
Theorem 2 . A tree with m-vertices has m-1 edges

Which graphs are trees?

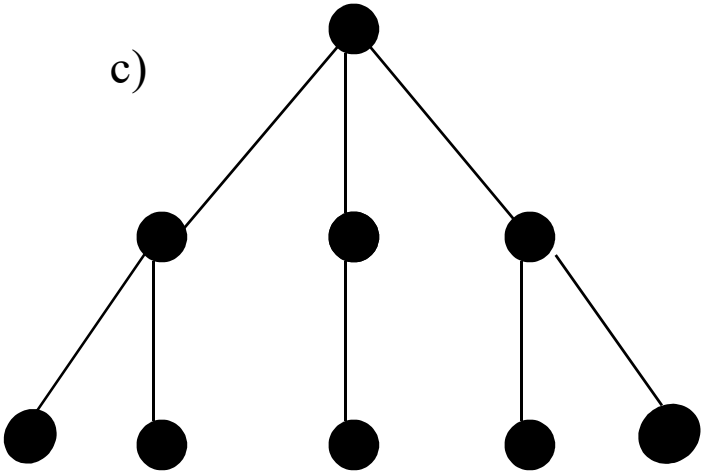
a)



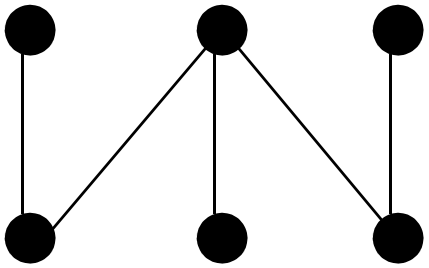
b)



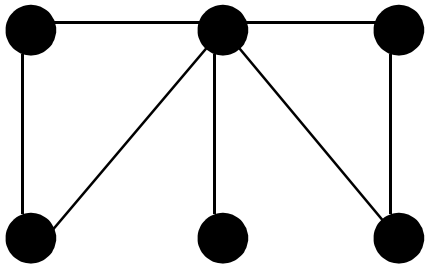
c)



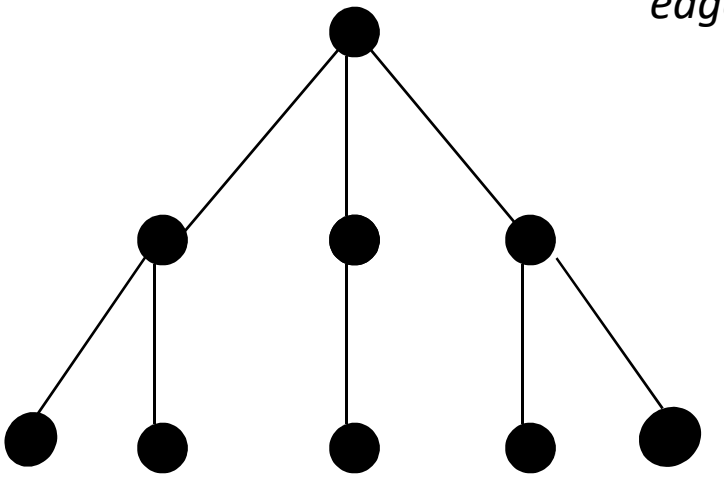
Solution



tree
vertices = 6
edges = 5



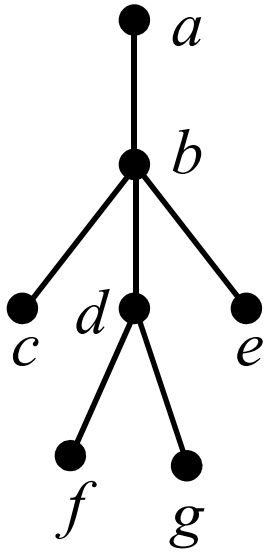
Not a tree
vertices = 6
edges = 7



tree
vertices = 9
edges = 8

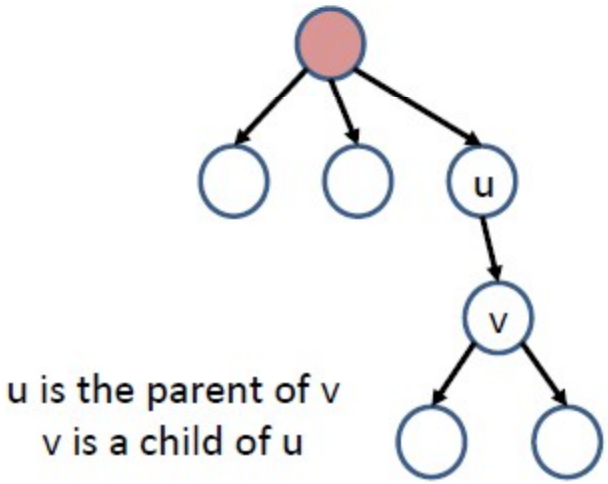
Rooted tree

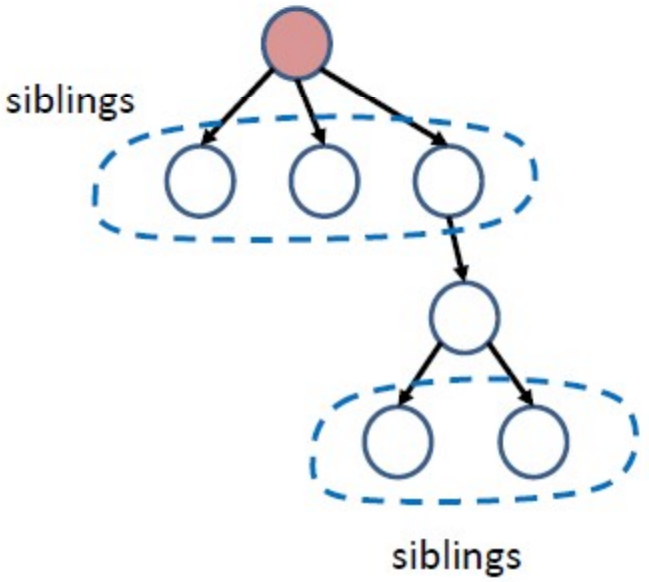
Definition 2. A **rooted tree** is a tree in which one vertex has been designed as the **root** and every edge is directed away from the root.



Rooted Tree - Terminologies

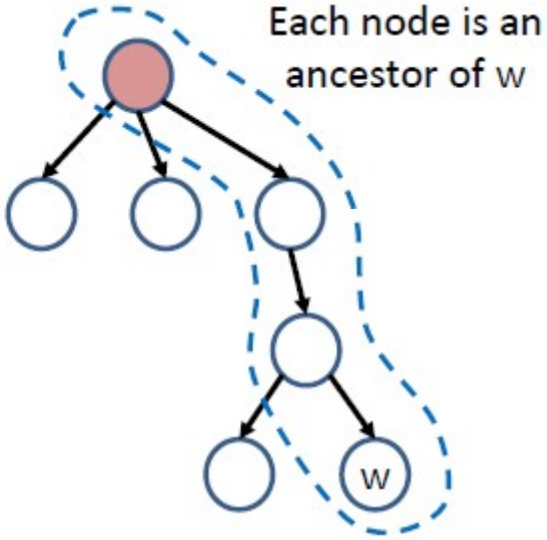
- Each edge is from a parent to a child
- Vertices with the same parent are siblings





Rooted Tree - Terminologies

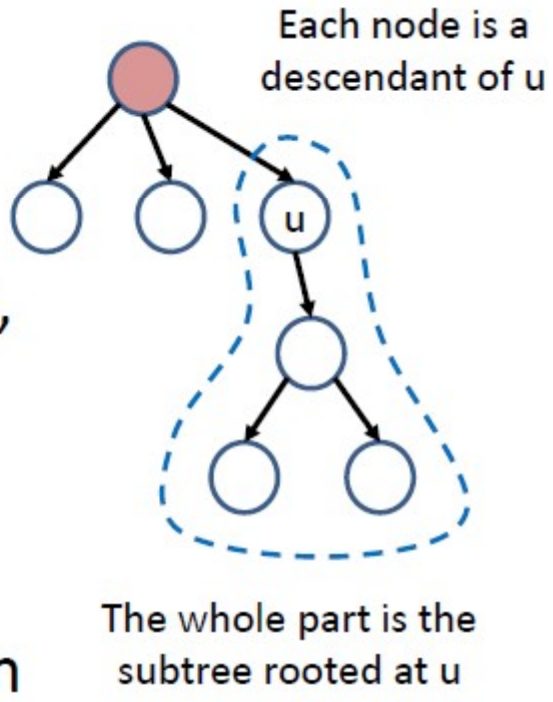
- The ancestors of a vertex w include all the nodes in the path from the root to w
- The proper ancestors of a vertex w are the ancestors of w, but excluding w



The whole part forms a path from root to w

Rooted Tree - Terminologies

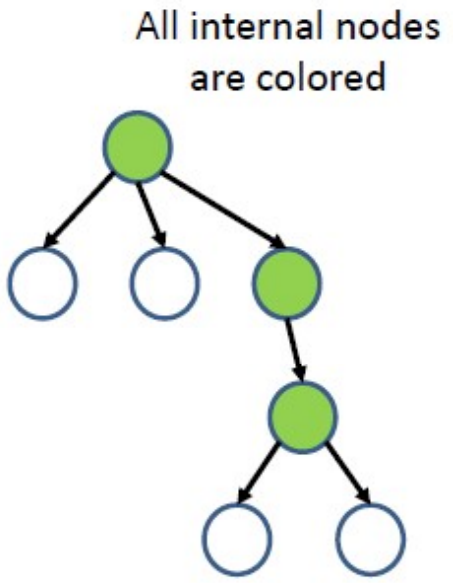
- The descendants of a vertex u include all the nodes that have u as its ancestor
- The proper descendants of a vertex u are the descendants of u, but excluding u
- The subtree rooted at u includes all the descendants of u, and all edges that connect between them



Rooted Tree - Terminologies

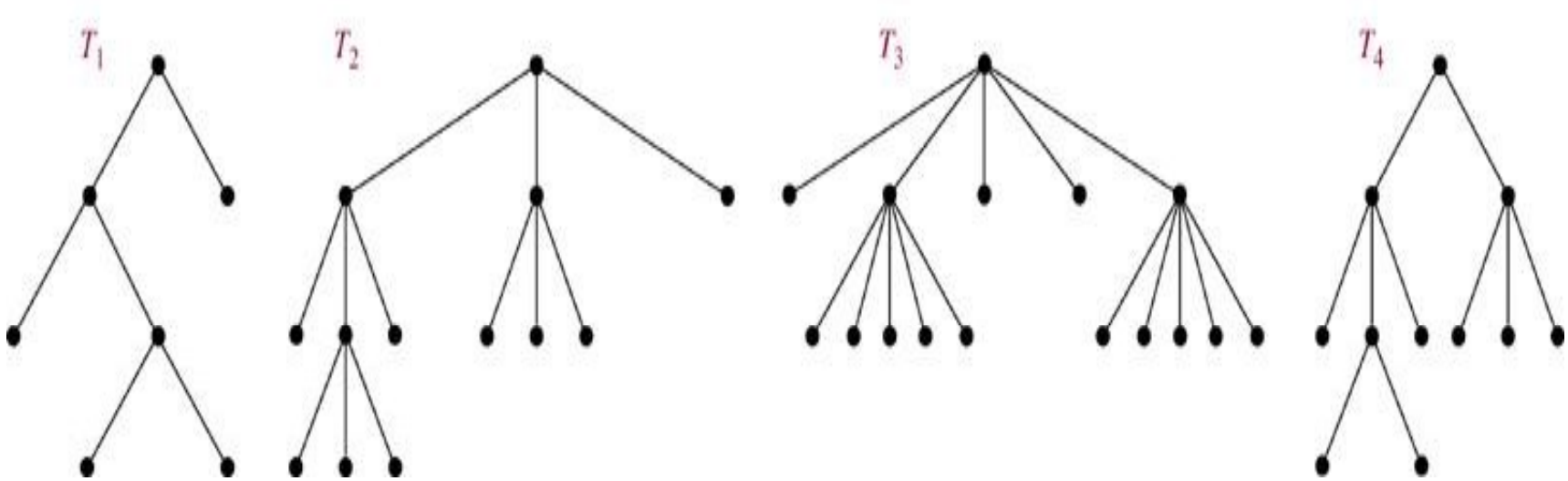
- Vertices with no children are called leaves ;

  Otherwise, they are called internal nodes
- If every internal node has no more than m children, the tree is called an m-ary tree
  - Further, if every internal node has exactly m children, the tree is a full m-ary tree



The tree is ternary (3-ary), but not full

Examples



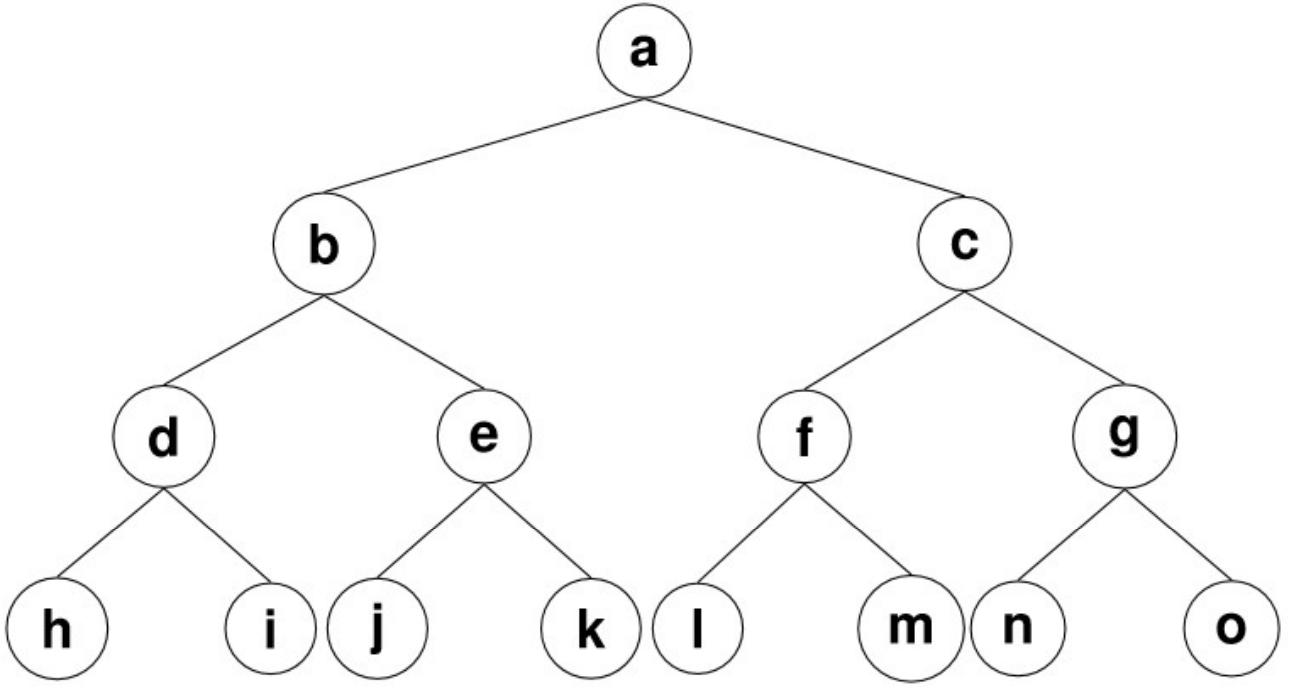
full binary tree

full 3-ary tree

full 5-ary tree

not full 3-ary tree

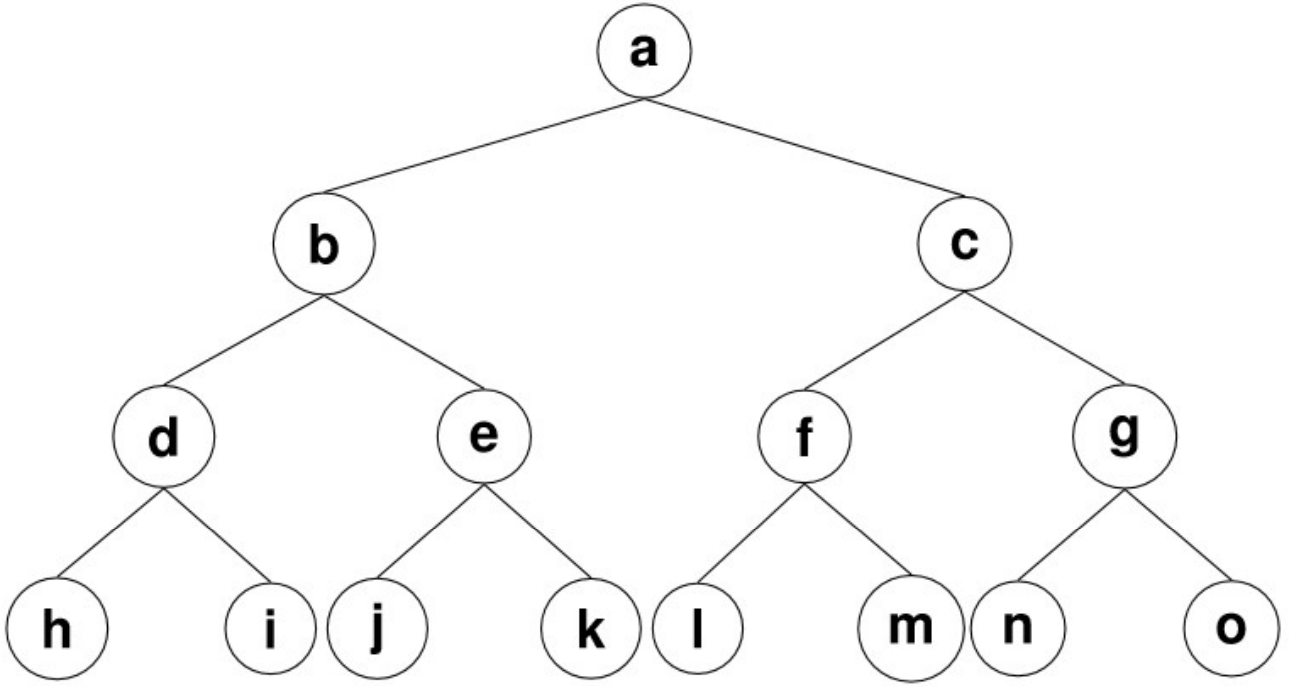
Exercise



Find:

- Ancestors of *g*
- Descendents of *g*
- Parent of *e*
- Children of *e*
- Sibling of *h*

Solution



Find:

- Ancestors of *g:* *c, a*
- Descendents of *g:* *n, o*
- Parent of *e:* *b*
- Children of *e:* *j, k*
- Sibling of *h:* *i*

Properties of Trees

- Theorem 2 : A tree with n nodes has n-1 edges
- Theorem 3 : A full m-ary tree with i internal vertices contains $n = mi + 1$ vertices.

Corollary: A full m-ary tree with n vertices contains $(n-1)/m$ internal vertices, and hence $n - (n-1)/m = ((m-1)n + 1)/m$ leaves

Corollary is a result in which the (usually short) proof relies heavily on a given theorem

Properties of Trees

Theorem 4 – A full *m*-ary tree with:

1. *n* vertices has $i = \frac{(n-1)}{m}$ internal vertices and

   $l = \frac{(m-1)n+1}{m}$ leaves

Properties of Trees

Theorem 4 – A full *m*-ary tree with:

2. *i* internal vertices has $n = mi + 1$ vertices and

   $l = (m-1)i + 1$ leaves

Cont.

Properties of Trees

Theorem – A full *m*-ary tree with

3. *l* leaves has $n = \frac{(ml-1)}{(m-1)}$ vertices and

$i = \frac{(l-1)}{(m-1)}$ internal vertices

Cont.

Example

Ex : Peter starts out a chain mail. Each person receiving the mail is asked to send it to four other people. Some people do this, and some don't

Now, there are 100 people who received the letter but did not send it out

Assuming no one receives more than one mail. How many people have sent the letter ?

Solution

- The chain letter can be represented using **4-ary** tree. The internal vertices correspond to people who sent out the letter, and the leaves correspond to people who did not send it out. Since 100 people did not send out the letter, the number of leaves in this rooted tree is, *l*=100. The number of people have seen the letter is *n=(4x100-1)/(4-1)=133*. The number of internal vertices is *133-100=33*, people sent the letter.

Exercise

- How many matches are played in a tennis tournament of 27 players?

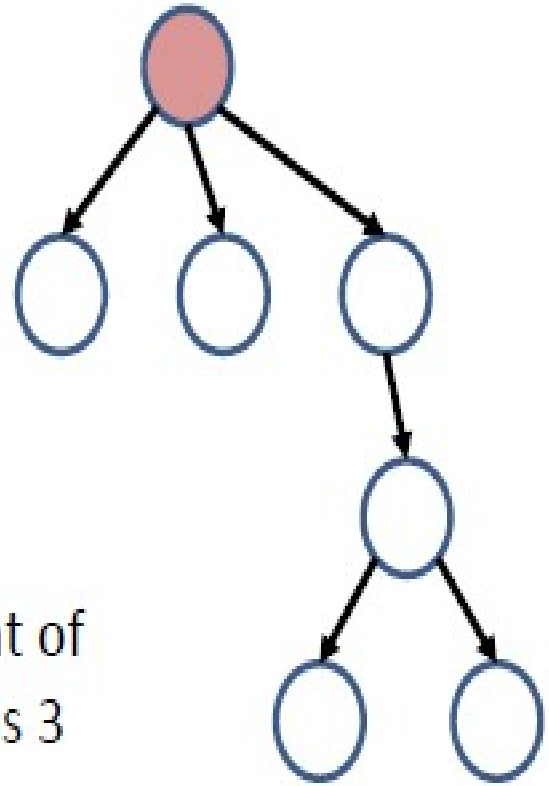
Exercise

Suppose 1000 people enter a chess tournament. Use a rooted tree model of the tournament to determine how many games must be played to determine a champion, if a player is eliminated after one loss and games are played until only one entrant has not lost. (Assume there are no ties.)

Properties of Trees

- The level of a vertex v in a rooted tree is the length of the unique path from the root to this vertex.
- The level of the root is defined to be zero.
- The height of a rooted tree is the maximum of the levels of vertices.

Example



Level 0

Level 1

Level 2

Level 3

This height of the tree is 3

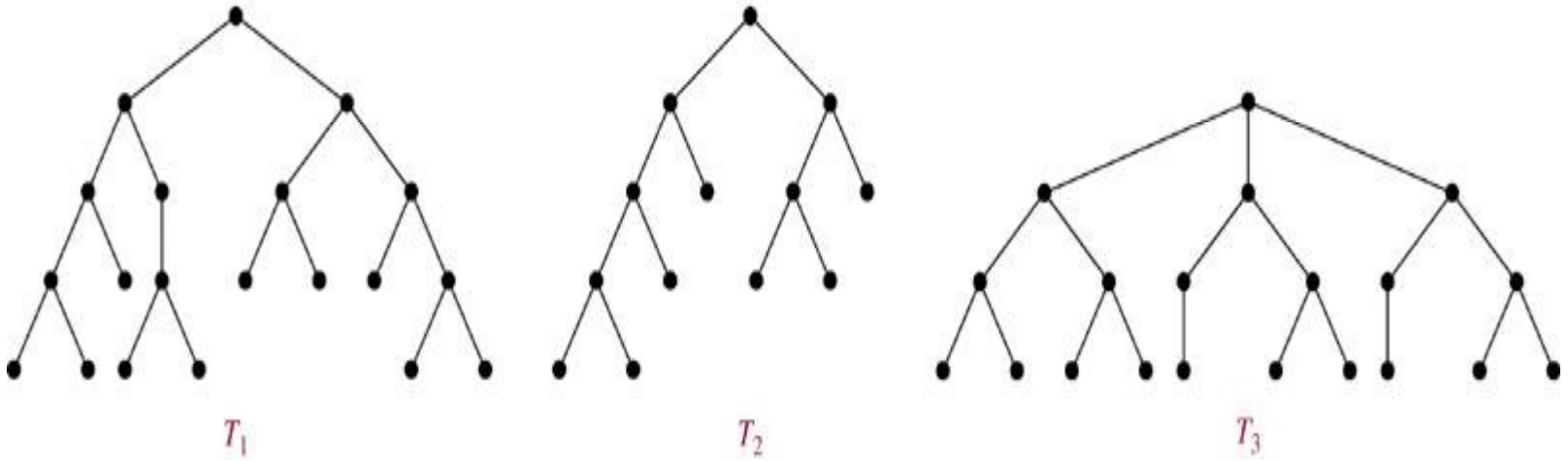
Properties of Trees

- **Definition:** A rooted m-ary tree of height h is balanced if all leaves are at levels h or $h-1$.

- **Theorem.** There are at most m^h leaves in an m-ary tree of height h.

Example

Which of the rooted trees shown below are balanced?



Sol. T_1, T_3